The Pythagorean Expectation, 45 Seasons Later

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Question

Bill James's Pythagorean expectation says that a baseball team's ratio of wins to losses is its ratio of runs scored to runs allowed raised to a power k:

$$\frac{W}{L} = \left(\frac{RS}{RA}\right)^{k},$$

with $k = 2$ in James's 1980 formulation and about 1.83 in later refinements. It is used to judge whether a team's record overstates or understates its quality. Does the formula still work on the 45 seasons played since it was published, and does the part of a record it does not explain carry over to the next season?

Result

The formula survives a long out-of-sample test. The exponent estimated on 1901–1980 is 1.847 and on 1981–2025 is 1.848. Applying the 1901–1980 fit to later seasons predicts a team's wins with an error of about 4 per 162 games, explaining 88% of the variation in wins. Wins above or below the formula do not persist: about 92% disappears the next season, and the formula's prediction is a better guide to next season's record than the actual record is. Figure 1 summarizes the three tests.

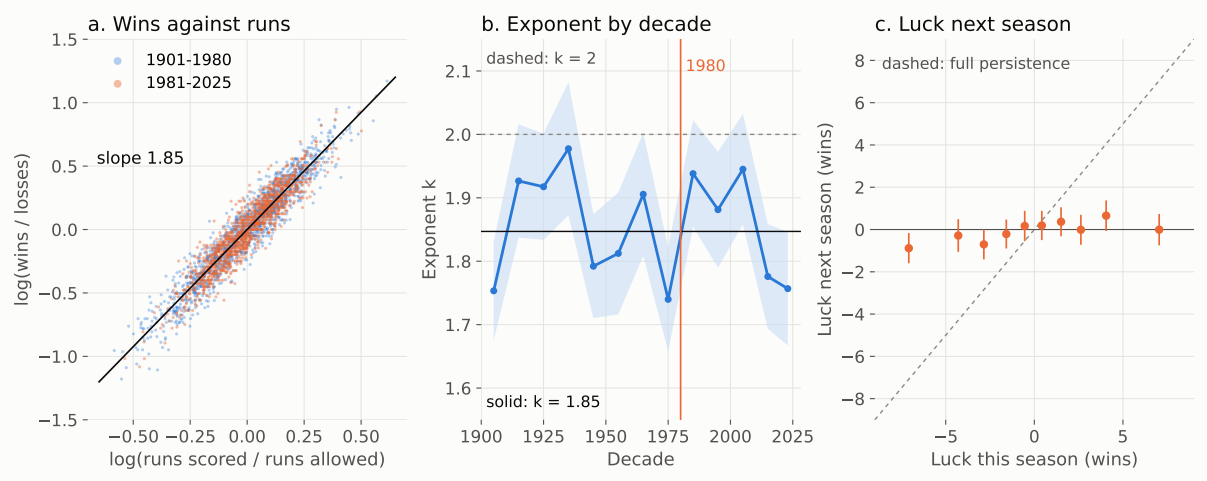


Figure 1: Pythagorean expectation, Major League Baseball, 1901–2025. Panel a: log of wins over losses against log of runs scored over runs allowed for 2,706 team-seasons; the line has slope 1.85. Panel b: exponent k estimated by decade, with bands of ±2 standard errors; dashed line at $k = 2$, solid line at the full-sample estimate, vertical line at 1980. Panel c: 1981–2024 team-seasons sorted into ten groups by wins above the formula ("luck"), against the group's mean luck in the next season, with ±2 standard errors; the dashed line is what full persistence would show.

Why it matters

Most empirical "laws" are tested in the sample that produced them. This one has a clean holdout: every season after 1980, none of which James could have seen, and it passes with an unchanged exponent. The practical lesson is about separating skill from noise. A record that beats its run differential by five or

more wins happens in about one season in four, and it is mostly noise; team quality is better read from runs than from wins. The same problem, separating a persistent component from a large transitory one, arises when judging fund managers, executives or forecasters from short records.

Data and design

Data. The Lahman Baseball Database, 2026 release (R package `Lahman` 14.0-0), team-season table. The sample is the American and National Leagues from 1901 to 2025: 2,706 team-seasons. The 2025 rows are complete (all teams 162 games; total wins equal total losses; total runs scored equal total runs allowed).

Exponent. k is the least-squares slope of $\log(W/L)$ on $\log(RS/RA)$ without an intercept, with heteroskedasticity-robust standard errors. The holdout is 1981–2025.

Accuracy. The predicted winning share is $RS^k/(RS^k + RA^k)$. Errors are differences between actual and predicted winning shares, scaled to 162 games, summarized by the root mean squared error.

Luck. "Luck" is actual minus predicted wins per 162 games with $k = 1.83$. Persistence is the correlation between a franchise's luck in consecutive seasons. To compare predictors of next season's winning share, it is regressed on this season's actual and predicted shares, separately and together.

	1901–1980	1981–2025
Exponent k	1.847	1.848
(standard error)	(0.016)	(0.020)
Prediction error, $k = 2$ (wins per 162 games)	4.47	4.16
Prediction error, $k = 1.83$	4.33	4.06
Prediction error, k fitted on 1901–1980	4.32	4.06
Standard deviation of wins per 162 games	14.65	11.63
Share of variation in wins explained, $k = 1.83$	0.91	0.88
Luck carried into next season (correlation)	0.04	0.08
(t-statistic)	(1.6)	(2.7)
Next season, weight on predicted share (both included)	0.59	0.40
Next season, weight on actual share (both included)	0.12	0.13

Table 1: Pythagorean expectation before and after publication. Luck rows use consecutive seasons of the same franchise (1,388 pairs for 1901–1979 and 1,262 for 1981–2024). The last two rows are coefficients from a regression of next season's winning share on this season's actual and predicted shares.

The exponent is the same before and after 1980 to the third decimal, and fixing it at 1.83 or at the 1901–1980 estimate gives the same out-of-sample error, 4.06 wins. James's original $k = 2$ is slightly worse. By decade the exponent ranges from 1.74 to 1.98, which is more variation than sampling error alone would produce ($\chi^2 = 45.9$ with 12 degrees of freedom), but it has no trend (-0.002 per decade). A refinement that lets the exponent rise with runs per game, $k = (\text{runs per game})^{0.287}$, does not improve the out-of-sample error (4.066 against 4.064 wins). Luck has a standard deviation of 4.2 wins. Its carry-over to the next season is 0.04 before 1980 and 0.08 after, which is statistically different from zero after 1980 but small: ten wins of luck predict less than one win of luck the following year. When this season's actual and predicted shares are both used to forecast next season, the predicted share receives about three times the weight.

Checks and caveats

Excluding the 60-game 2020 season leaves the post-1980 exponent at 1.847. Strike-shortened seasons (1981, 1994, 1995) are included; scaling by games played puts them on a common footing. The data are runs and decisions only, so the note cannot say what drives the small post-1980 persistence in luck; strong bullpens, which win close games consistently, are one candidate. The relationship is descriptive and holds on average; a single team can deviate by ten wins or more.

Relation to the literature

James introduced the formula in his *Baseball Abstract* around 1980. Davenport and Woolner (1999) proposed an exponent that varies with the scoring environment, and Miller (2007) derived the formula

from Weibull-distributed run scoring. That luck regresses and that runs predict better than wins are long-standing points in baseball analysis. This note contributes a clean pre- and post-publication comparison on 1901–2025 data with the three tests side by side; it does not claim a new finding.

References

[1] Davenport, C., and Woolner, K. (1999). Revisiting the Pythagorean Theorem: Putting Bill James' Pythagorean Theorem to the Test. *Baseball Prospectus*.

[2] Friendly, M., Dalzell, C., Monkman, M., and Murphy, D. (2026). *Lahman: Sean 'Lahman' Baseball Database*. R package version 14.0-0.

[3] James, B. (1980). *The Bill James Baseball Abstract*. Self-published.

[4] Miller, S. J. (2007). A Derivation of the Pythagorean Won-Loss Formula in Baseball. *Chance* 20(1), 40–48.

Data: Lahman Baseball Database (public). Drafted with AI assistance and checked before publication.